VICTOR OLUWATOBILOBA OGUNDIPE

Full-Stack & AI Systems Software Engineer

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PROFESSIONAL SUMMARY

Full-Stack & AI Systems Engineer (B.Eng. Computer Engineering) experienced in architecting end-to-end web platforms, evidence-grounded LLM & scholarly-graph pipelines, fintech/commerce payment workflows, and fault-tolerant network and edge systems. Proven track record shipping TypeScript/React, Python/FastAPI, Node.js/Express, and C++/FreeRTOS software with rigorous security guards and automated verification.

TECHNICAL SKILLS

Languages:	TypeScript, Python, JavaScript (ES2024+), C / C++, SQL, HTML5, CSS3
Frontend & Graphics:	React 18/19, Next.js 16, Vite, Tailwind CSS, Three.js (WebGL), HTML5 Canvas, Zustand, Leaflet, Progressive Web Apps (PWA)
Backend, APIs & Payments:	FastAPI (ASGI), Node.js, Express, Square API (Web Payments SDK, Card Auth/Capture/Void), REST APIs, WebSockets, OAuth 2.0 (PKCE), JWT (HS256)
AI, Data & Geospatial:	Gemini API (768-dim Embeddings & Flash), Claude API, OpenAlex & ROR v2 APIs, PyMuPDF4LLM, OSRM, Google Route Matrix, Pandas, NumPy, Scikit-learn, Selenium
Data Stores, Infra & Security:	Cloud Firestore, Firebase RTDB, SQLite (WAL), Cloudinary, Nginx (CIDR Perimeter / Reverse Proxy), Docker Compose, AES-256-GCM, Proof-of-Work & Request Signing
Embedded, Testing & Tooling:	ESP32-S3, FreeRTOS, I²C/SPI/UART, Vitest, fast-check (Property-Based Testing), Pytest, Node Test Runner, Git, GitHub Actions, Vercel, Render, Linux

SELECTED SOFTWARE ENGINEERING PROJECTS

ThesisBridge — Evidence-Grounded Academic Intelligence & Outreach Platform

2026 – Present

`React 18 · TypeScript · FastAPI · Python 3.11+ · Firestore · OpenAlex · Gemini & Claude APIs`

github.com/victobiloba/ThesisBridge

- Architected a full-stack matchmaking engine resolving universities to canonical ROR v2 IDs and indexing faculty across OpenAlex (250M+ works), enforcing a strict source-backed-or-empty contract and an ORCID/byline career-stage gate to eliminate hallucinated advisors.
- Built a dual-document ingestion pipeline (`pymupdf4llm`) parsing thesis PDFs and CVs into a 4-category skill taxonomy, ranking research fit with 768-dim Gemini embeddings, Gemini Flash / Claude fallback evaluation, and verified NSF/NIH RePORTER grant attribution.
- Engineered default-deny HS256 JWT session auth, an AES-256-GCM token vault, sliding-window IP rate limiting, minimal-scope OAuth 2.0 PKCE Sign-In, a rubric-scored Statement of Purpose Studio (DOCX/PDF export), and an outreach CRM with A/B tone telemetry.

Farm Pay — Multi-Tenant E-Commerce Storefront, Wholesale & Farmer Terminal

2026

`Node.js 22 · Express · Square API · JavaScript · QR Pickup Scanner · node:test`

- Built a full-stack commerce store and merchant portal on the Square API supporting retail checkout (Apple Pay, Google Pay, cards), access-gated wholesale ordering (case pricing, configurable $200k caps, delivery/pickup routing), and direct browser-to-Square tokenization.
- Engineered a `/farmer` phone-order terminal implementing two-step card-on-file authorization holds with deferred capture upon packing, voiding, signed payment links, and QR pickup verification, backed by progressive-enhancement shared pricing modules and mock-Square automated tests.

MedRoute — Capability-Based Emergency Medical Routing Platform

2026

`TypeScript Monorepo · React 19 PWA · Node.js · Leaflet · OSRM · Vitest · fast-check`

- Developed an Expected Time to Definitive Care (ETDC) routing engine (`POST /v2/routing/evaluate`) matching patients to facilities verified to treat their emergency rather than merely the nearest clinic, featuring stabilize-first triage, an F0–F4 fallback ladder, capacity leases, and idempotent retries.

- Combined self-hosted OSRM (Nigeria OSM road graph), circuit-breaker-guarded Google Route Matrix live-traffic refinement with hysteresis, and Haversine `L1` fallback; built a 5-language React 19 Citizen PWA and verified core invariants (non-emptiness, determinism, monotonicity) with `fast-check`.

Abuja Story — 3D Browser Life Simulation & Anti-Automation Security Engine

2026

`Next.js 16` · `React 19` · `Three.js` · `TypeScript` · `Zustand` · `Python (OSM)` · `Firebase`

github.com/victobiloba/abjlife

- Built a browser 3D city simulation generated from real OpenStreetMap Abuja roads, waterways, and landmarks via a Python geometry pipeline (`build-abuja-map.py`), featuring graph-routed rides (`roadGraph.ts`), procedural audio synthesis, encrypted messaging, and multiplayer social systems.
- Engineered a custom defense-in-depth anti-bot guard (`lib/guard/`): Proof-of-Work `/api/handshake` challenges, WebDriver/CDP headless detection, ephemeral in-memory request signing keys preventing `curl` cookie replay, synthetic DOM event rejection, and bundle obfuscation.

NetShare LAN — Subnet-Isolated File Sharing & Media Streaming Infrastructure

2026

`Python` · `FastAPI` · `Nginx` · `WebSockets` · `SQLite (WAL)` · `Docker Compose` · `Pytest`

- Engineered a two-layer intranet file-sharing system pairing an Nginx RFC1918 CIDR reverse-proxy perimeter with FastAPI middleware that dynamically introspects host NICs to enforce strict same-subnet (`STRICT_SAME_SUBNET`) access and audit blocked WAN traffic.
- Implemented a 5 MB chunked multi-gigabyte upload pipeline with SHA-256 verification, HTTP 206 Partial Content byte-range video/audio streaming, real-time WebSocket LAN peer presence, and SQLite WAL multi-user authentication.

Codebase DNA — Interactive GitHub Repository Organism & Dead-Code Visualizer

2026

`JavaScript (ES Modules)` · `HTML5 Canvas` · `Static Import Graph Analysis` · `Node.js`

- Created a zero-dependency static-analysis and physics-driven Canvas visualizer mapping GitHub repos into living organisms (modules as organs, files across 15+ languages as cells, imports as veins, high fan-in hotspots as stress, and unreferenced files as scar tissue) with framework-route dead-code heuristics and Markdown/PNG export.

HerdSense — Edge-ML Wearable Biosensing & Cloud Telemetry System

Mar 2026 – Aug 2026

`C++` · `ESP32-S3` · `FreeRTOS` · `LittleFS` · `Firebase RTDB` · `EasyEDA PCB` · `Web Dashboard`

github.com/victobiloba/HerdSense

- Built end-to-end multi-modal biosensing firmware (4 concurrent FreeRTOS tasks, 3 mutexes) sampling PPG, IR thermometry, 25 Hz IMU, and ToF contact-gating across a dual I²C bus with on-device C++ DFT spectral extraction, 3-day LittleFS flash buffering, Firebase RTDB bulk resync, and a web dashboard with CSV export & Telegram alerts.

WORK EXPERIENCE & OPEN SOURCE

Freelance Software Automation Engineer — Independent Contract

Sep 2024 – Feb 2025

Remote, Nigeria

`Python` · `Selenium WebDriver` · `Pandas` · `NumPy`

- Engineered and deployed a Python/Selenium browser automation pipeline handling authenticated session management, dynamic DOM synchronization, and structured extraction, validating outputs with Pandas and NumPy across the full contract lifecycle.

Open-Source Contributor — `sciencecorp/synapse-python`

Oct 2026

Science Corp · Remote

`Python` · `gRPC Simulator` · `BCI Device Client`

- Diagnosed and patched a `SpikeSource` simulator startup crash in Science Corp's Python BCI device client (`synapse-python`).

EDUCATION & PUBLICATIONS

B.Eng. Computer Engineering — Federal University of Technology Akure (FUTA)

Oct 2020 – Expected Nov 2026

Second Class Upper Honours · Akure, Nigeria

DSA · Networks · Embedded · DSP

- **IEEE Publication (2026):** Ogundipe, V.O., & Adedokun, A. *"Development of an IoT-based Wearable Device for Real-Time Health and Activities Monitoring of Native Fulani Cattle."* 2026 IEEE NIGERCON, Lagos *(Conditionally accepted)*.
- **Machine Learning Training:** Applied Machine Learning Fellow, *AI Federax* (2024–2025), a year-long applied ML programme.